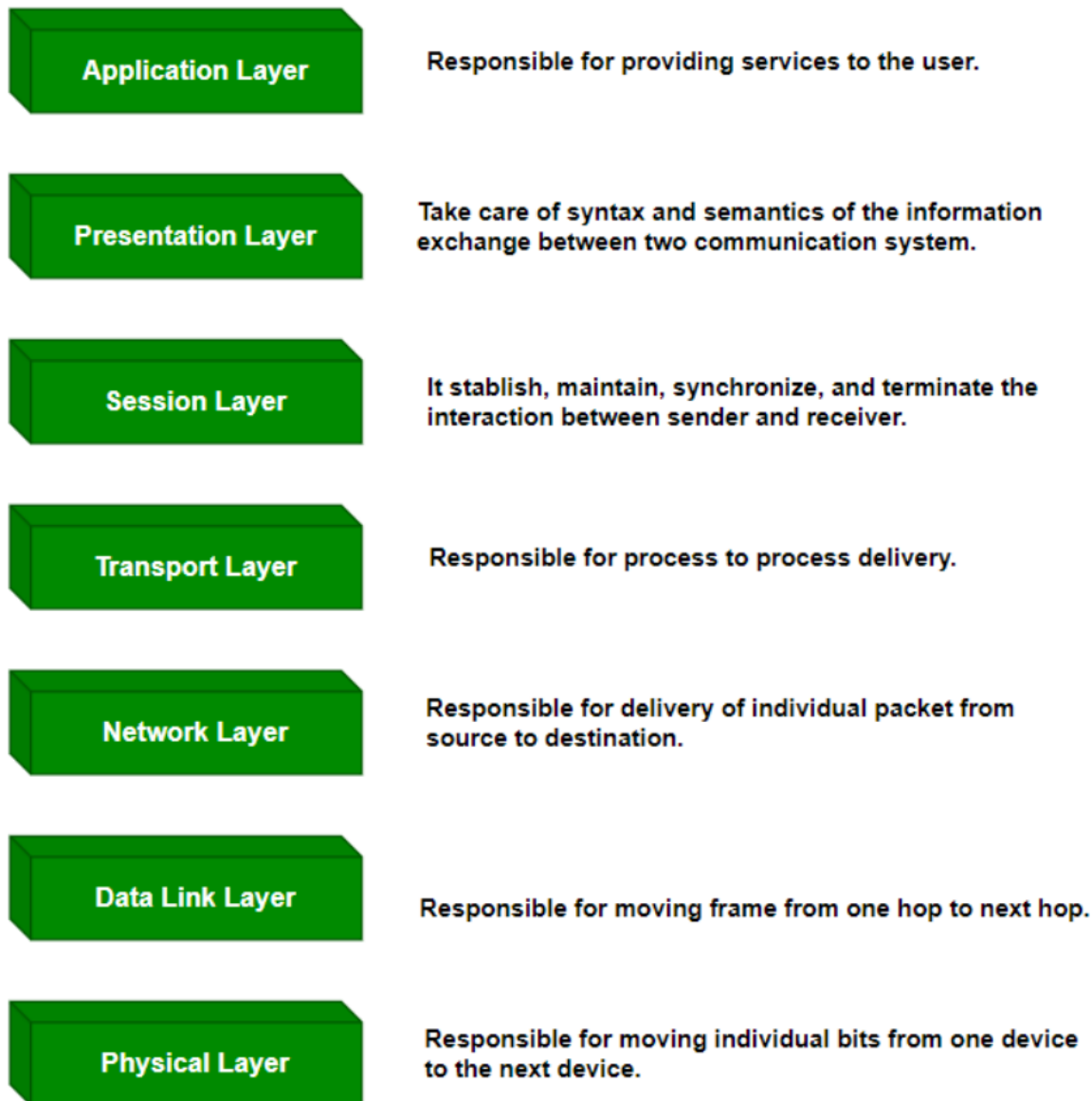


1. Layers of OSI Reference Model

Ans: OSI stands for Open System Interconnection



2. Explain TCP/IP Reference Model.

Ans: 4 layers of TCP/IP Model

1. Host-to-Network Layer /Network access layer
2. Internet Layer
3. Transport Layer
4. Application Layer

3. Differences between ISO-OSI model and TCP/IP model

ISO-OSI model	TCP/IP model
OSI stands for Open Systems Interconnection	TCP/IP stands for Transmission Control Protocol/Internet Protocol
It has 7 layers	It has 4 layers
It is low in usage	It is mostly used
An OSI Model is a reference model, based on which a network is created.	The TCP/IP is the implementation of the OSI Model.
It is protocol-independent.	It is protocol-dependent.
It follows a vertical approach.	It follows a horizontal approach.
Replacement of tools and changes can easily be done in this model	Replacing the tools is not easy as it is in OSI Model
It is less reliable than TCP/IP Model	It is more reliable than OSI Model
In this model, the session and presentation layers are separated, i.e., both the layers are different.	In this model, the session and presentation layer are not different layers. Both layers are included in the application layer.

4. List the various Transmission Modes in Computer Networks

Ans: The way in which data is transmitted from one device to another device is known as transmission mode/communication mode.

a. Simplex mode : The communication is unidirectional

b. Half-Duplex Mode: each station can both transmit and receive, but not at the same time.

Example: Walkie-talkie

c. Full-Duplex Mode : The communication is bi-directional, i.e., the data flow in both the directions. Both stations can transmit and receive simultaneously.

5. Explain Network Topology

Ans : Network Topology is the pattern used to arrange the nodes of network (describe the way computers are connected).

Eg: Bus, mesh, star, tree, ring, hybrid

6. Explain Transmission Media

Ans: Transmission media is a communication channel that carries the information from the sender to the receiver.

a. Guided Media : Guided Media is also referred to as Wired or Bounded transmission media.

Eg : Twisted Pair Cable , Coaxial Cable , Optical Fiber Cable

b. Unguided Media : It is also referred to as Wireless or Unbounded transmission media.

Eg: Radio Waves, Microwaves, Infrared

7. List any 6 Difference between Radio wave, Microwave, and Infrared waves

Ans:

Sr. No.	Basis	Radiowave	Microwave	Infrared wave
1.	Direction	These are omni-directional in nature.	These are unidirectional in nature.	These are unidirectional in nature.
2.	Penetration	At low frequency, they can penetrate through solid objects and walls but high frequency they bounce off the obstacle.	At low frequency, they can penetrate through solid objects and walls. at high frequency, they cannot penetrate.	They cannot penetrate through any solid object and walls.
3.	Frequency range	Frequency range: 3 KHz to 1GHz.	Frequency range: 1 GHz to 300 GHz.	Frequency range: 300 GHz to 400 THz.
4.	Security	These offers poor security.	These offers medium security.	These offers high security.
5.	Attenuation	Attenuation is high.	Attenuation is variable.	Attenuation is low.
6.	Government License	Some frequencies in the radio-waves require government license to use these.	Some frequencies in the microwaves require government license to use these.	There is no need of government license to use these waves.
7.	Usage Cost	Setup and usage Cost is moderate.	Setup and usage Cost is high.	Usage Cost is very less.
8.	Communication	These are used in long distance communication.	These are used in long distance communication.	These are not used in long distance communication.

8.

8. Explain Bandwidth-Delay Product

Ans: It is the product of the bandwidth-delay product. The bandwidth-delay product defines the number of bits that can fill the link.

Eg: Consider that the link capacity of a channel is 512 Kbps and round-trip delay time is 1000ms.

The bandwidth delay product = $512 \times 10^3 \text{ bits/sec} \times 1000 \times 10^{-3} \text{ sec}$

= 512,000 bits = 64,000 bytes = 62.5 KB

9. Explain Framing in datalink layer

Ans: Framing is the major function of datalink layer. Data-link layer takes the packets from the Network Layer and encapsulates them into frames.

Methods of variable Framing

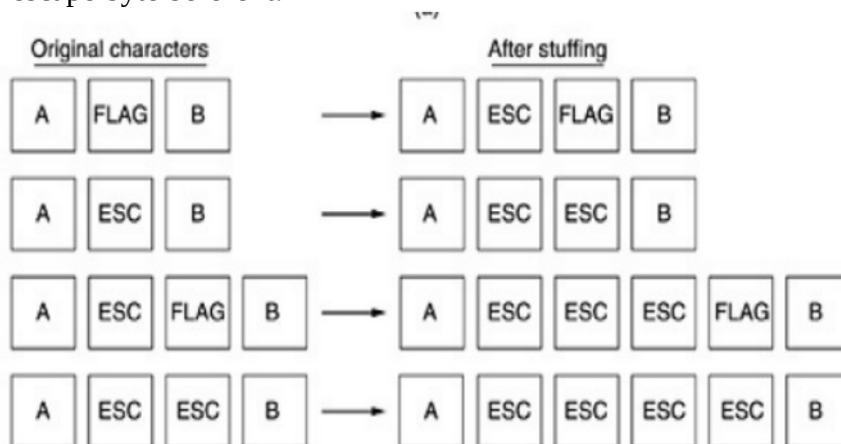
- 1. Character count.
- 2. Flag bytes with byte stuffing.
- 3. Starting and ending flags, with bit stuffing.

10. Explain Flag bytes with byte stuffing.

Ans : insert a special escape byte (ESC) just before each "accidental" flag byte in the data.

This technique is called byte stuffing or character stuffing.

Thus, a framing flag byte can be distinguished from one in the data by the absence or presence of an escape byte before it.



11. The original data : A B ESC C ESC FLAG FLAG D is given. What is the output after stuffing?

ANS:

After stuffing, we get A B ESC ESC C ESC ESC ESC FLAG ESC FLAG D

12. Explain Bit stuffing

Ans: In the data, a 0 is automatically inserted after 5 consecutive 1's.

(a) 0 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 0

a) Original data.

(b) 0 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0 1 0 0 1 0

b) Data as they appear on the transmission line

Stuffed bits

13. Sliding window protocols

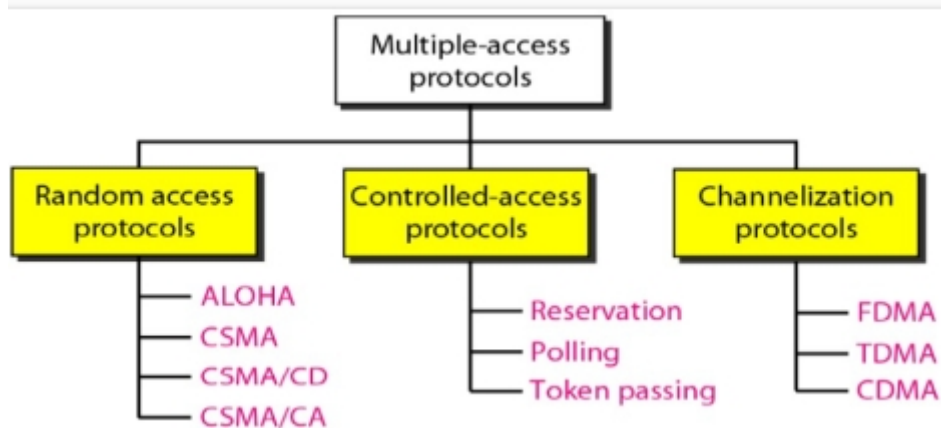
Ans:

a. Stop-and-Wait Protocol : Source send a packet. It sends the next packet only after receiving the acknowledgement from destination.

- b. Go-back-N protocol : If the ACK of a frame is not received within the time period, all frames starting from that frame are retransmitted
- c. Selective Repeat : Only the erroneous or lost frames are retransmitted while good frames are received and buffered. Sender individually transmits frames that have timed out.
If a frame is corrupted in transit, a NAK is returned and that frame is resend out of sequence.

14.Explain Multiple Access Protocols

Ans: Multiple Access Protocols are protocols used in computer networks to control how data is transmitted when multiple devices are trying to communicate over the same network.



15. Difference between switch and bridge

Ans:

S.NO.	Switch	Bridge
1.	It is a device which is responsible for channeling the data that is coming into the various input ports to a particular output port which will further take the data to the desired destination.	It is basically a device which is responsible for dividing a single network into various network segments.
2.	A switch can have a lot of ports.	A bridge can have 2 or 4 ports only.
3.	The switch performs the packet forwarding by using hardwares such as ASICS hence, it is hardware based.	The bridge performs the packet forwarding by using softwares so it is software based.
4.	The switching method in case of a switch can thus be store, forward, fragment free or cut through.	The switching method in case of a bridge is store and forward.
5.	The task of error checking is performed by a switch.	A bridge cannot perform the error checking.
6.	A switch has buffers.	A bridge may not have a buffer.

17. What is a gateway?

Ans:A gateway is a network node or device that connects two networks that use different transmission protocols.

- Gateways play an important role in connecting two networks.

- It works as the entry-exit point for a network because all traffic that passes across the networks must pass through the gateway.
- A network gateway is a device that connects different networks by translating messages from one protocol into another protocol.

18. Compare Virtual Circuit Networks and Datagram Networks

Ans:

Criteria	Virtual Circuit Networks	Datagram Networks
Connection Establishment	Prior to data transmission, a connection is established between sender and receiver.	No connection setup is required.
Routing	Routing decisions are made once during connection setup and remain fixed throughout the duration of the connection.	Routing decisions are made independently for each packet and can vary based on network conditions.
Flow Control	Uses explicit flow control, where the sender adjusts its rate of transmission based on feedback from the receiver.	Uses implicit flow control, where the sender assumes a certain level of available bandwidth and sends packets accordingly.
Congestion Control	Uses end-to-end congestion control, where the sender adjusts its rate of transmission based on feedback from the network.	Uses network-assisted congestion control, where routers monitor network conditions and may drop packets or send congestion signals to the sender.
Error Control	Provides reliable delivery of packets by detecting and retransmitting lost or corrupted packets.	Provides unreliable delivery of packets and does not guarantee delivery or correctness.
Overhead	Requires less overhead per packet because connection setup and state maintenance are done only once.	Requires more overhead per packet because each packet contains information about its destination address and other routing information.

19. List the Properties of Routing algorithms

Ans:

- Optimality – capability of routing algorithm to select the best route
- Simplicity and low overhead -With increasing complexity of the routing algorithms the overhead also increases
- Robustness – means they should perform correctly in the face of unusual circumstances such as hardware failure
- Stability -The routing algorithms should be stable under all possible circumstances.
- Flexibility – they should quickly & accurately adapt to a variety of network circumstances
- Fairness - Every node connected to the network should get a fair chance of transmitting their packets.

20. State the Optimality Principle in routing.

Ans: The optimality principle states that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route.

21. Explain Flooding

Ans: Flooding is the static routing algorithm.

In this algorithm, every incoming packet is sent on all outgoing lines except the line on which it has arrived

Advantages of Flooding

- It is very simple to setup and implement, since a router may know only its neighbours.

- It is extremely robust. Even in case of malfunctioning of a large number routers, the packets find a way to reach the destination.
- All nodes which are directly or indirectly connected are visited. So, there are no chances for any node to be left out.
- The shortest path is always chosen by flooding.

Limitations of Flooding

- Flooding tends to create an infinite number of duplicate data packets
- The network may be clogged with unwanted and duplicate data packets.
- It is wasteful if a single destination needs the packet, since it delivers the data packet to all nodes irrespective of the destination.

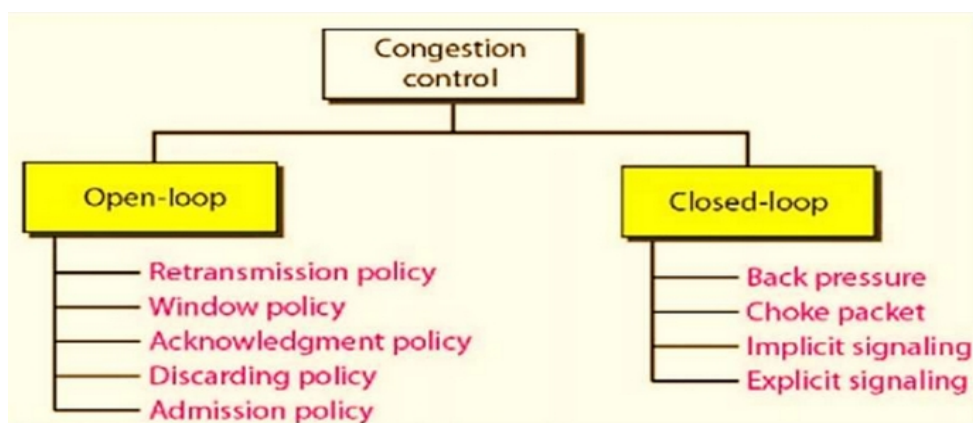
22. Comparison between Distance Vector Routing and Link State Routing

Ans:

<u>Distance Vector Routing</u>	<u>Link State Routing</u>
Each router periodically share its knowledge about the entire network with its neighbours	Router share the knowledge of only their neighbouring routers to entire network
Updates full routing table	Updates only the link state
Bandwidth required is less due to local sharing, small packets and no flooding.	Bandwidth required is more due to flooding and sending of large link state packets.
Make use of Bellman Ford Algorithm. .	Make use of Dijkstra's algorithm.
Simple to implement and manage	Complex and requires trained network administrator
Traffic is less.	Traffic is more.
Converges slowly	Converges faster.
Count of infinity problem.	No count of infinity problem.

23. Explain Congestion control

Ans: It refers to techniques and mechanisms that can either prevent congestion, before it happens, or remove congestion, after it has happened.



24. What is the difference between open-loop and closed loop congestion control

Ans: In open-loop congestion control, policies are applied to prevent congestion before it happens. Closed-loop congestion control mechanisms try to alleviate congestion after it happens.

25. Differentiate Backpressure and chokepacket.

Ans:

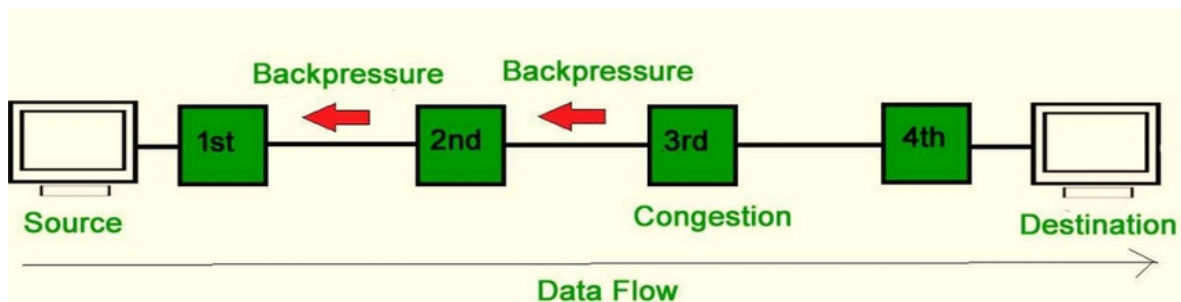
1. Backpressure

Node 3 in the figure has more input data than it can handle. It drops some packets in its input buffer and informs node 2 to slow down.

Node 2, in turn, may be congested because it is slowing down the output flow of data. If node 2 is congested, it informs node 1 to slow down, which in turn may create congestion in node 1.

If so, node 1 informs the source of data to slow down. This, in time, alleviates the congestion.

The pressure on node 3 is moved backward to the source to remove the congestion.



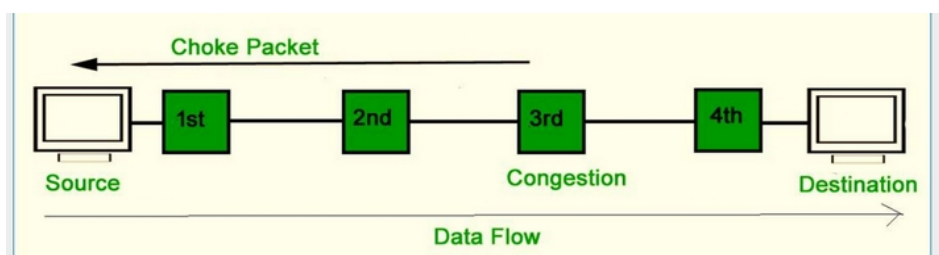
2. Choke Packet

A choke packet is a packet sent by a node to the source to inform it of congestion.

In backpressure, the warning is from one node to its upstream node, although the warning may eventually reach the source station.

In the choke packet method, the warning is from the router, which has encountered congestion, to the source station directly.

The intermediate nodes through which the packet has travelled are not warned.



26. List the techniques for achieving good quality of service in a network

ans:

- **Scheduling**
 - FIFO
 - Priority scheduling
 - Weighted Fair Queuing
 - **Traffic shaping**
 - Leaky bucket
 - token bucket
 - **Resource reservation**
 - **Admission control**
-

27. Explain the need of Internet Control Message Protocol (ICMP)

Ans: Internetworking protocol (IP) does not have an inbuilt mechanism for sending error and control messages. Ie, The IP protocol has no error-reporting or error-correcting mechanism. The IP protocol also lacks a mechanism for host and management queries.

So, IP depends on Internet Control Message Protocol(ICMP) to provide an error control.

–It is used for 1. reporting errors and 2.management queries.

–ICMP is a supporting protocol and is used by network devices like routers for sending error messages and operations information.

28. Explain the two categories of ICMP messages

Ans:

Category	Type	Message
Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply

Eg: **Destination unreachable** : The destination unreachable error occurs when the packet does not reach the destination. Suppose the sender sends the message, but the message does not reach the destination, then the intermediate router reports to the sender that the destination is unreachable.

Echo-request and echo-reply message :

- A router or a host can send an echo-request message.
 - It is used to ping a message to another host that "Are you alive".
 - If the other host is alive, then it sends the echo-reply message.
-

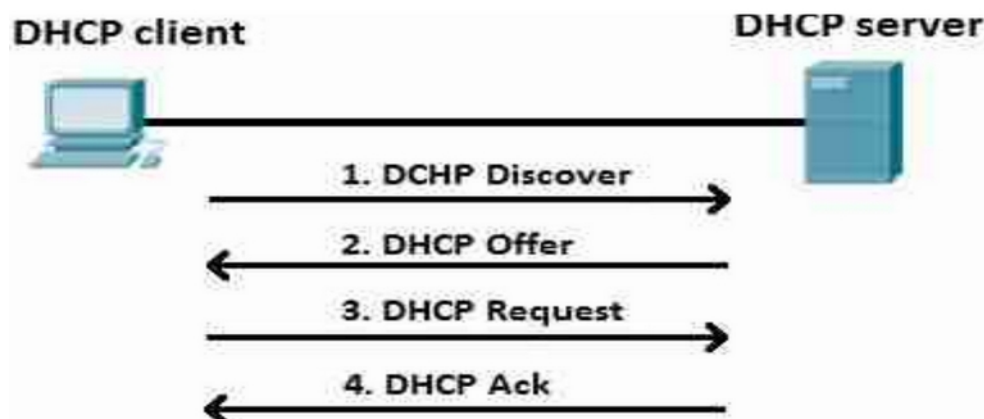
29. Differentiate ARP and RARP

ARP	RARP
ARP stands for Address Resolution Protocol.	RARP stands for Reverse Address Resolution Protocol.
A protocol used to map an IP address to a physical (MAC) address	A protocol used to map a physical (MAC) address to an IP address
Host1 broadcasts its IP address and requests a MAC address, and the host2 responds with the corresponding MAC address	Client broadcasts its MAC address and requests an IP address, and the server responds with the corresponding IP address
Through ARP, (32-bit) IP address mapped into (48-bit) MAC address.	Through RARP, (48-bit) MAC address of 48 bits mapped into (32-bit) IP address.
Request 1 and Reply 2	Request 3 and Reply 4
The primary use of the ARP reply is to update the ARP table.	The primary use of the RARP reply is to configure the local host's IP address.
Widely used in modern networks to resolve IP addresses to MAC addresses	Rarely used in modern networks as most devices have a pre-assigned IP address
The ARP table is maintained or managed by the local host.	The RARP table is maintained by the RARP Server.

30. Explain the purpose of Dynamic Host Configuration Protocol (DHCP)

Ans: it is a network management protocol used to 'dynamically' assign an IP address to any device, or node, on a network, so they can communicate using IP (Internet Protocol).

Working of DHCP



1. DHCP Discover Message:

- This message is generated by the Client host in order to discover if there is any DHCP server/servers are present in a network or not.
- This message is broadcasted to all devices present in a network to find the DHCP server.

2. DHCP Offers A Message:

- The server will respond to the host in this message specifying the unleased IP address and other TCP configuration information.

3. DHCP Request Message:

- When a client receives an offer message, it responds by broadcasting a DHCP request message.
- The client will produce a gratuitous ARP in order to find if there is any other host present in the network with the same IP address.

– If there is no reply from another host, then there is no host with the same TCP configuration in the network and the message is broadcasted to the server showing the acceptance of the IP address.

4. DHCP Acknowledgment Message:

–In response to the request message received, the server will make an entry with a specified client ID and bind the IP address offered with lease time.

–Now, the client will have the IP address provided by the server.

31. What is the maximum number of IP addresses that can be assigned to hosts on a local subnet that uses the 255.255.255.192 subnet mask?

Ans:

Given subnet mask=255.255.255.192

Ie, 11111111 11111111 11111111 11000000

Thus, maximum no.of IP addresses that can be assigned to hosts = $2^{\text{no.of host bits}} - 2 = 2^6 - 2 = 64 - 2 = 62$

32. Purpose of subnetting.

Ans: Subnetting is a process of dividing a single large network into multiple sub networks.

Uses of Subnetting

- Maintenance is easier for smaller networks. For example, if we consider a class A address, the possible number of hosts is 224 for each network, it is obvious that it is difficult to maintain such a huge number of hosts, but it would be quite easier to maintain if we divide the network into small parts.
- Subnetting helps in organizing the network in an efficient way which helps in expanding the technology for large firms and companies.
- Reduces traffic and maintain order and efficiency.
- Reduces network complexity
- Subnetting enhances the network's overall performance by removing redundant traffic.
- Increases network security .
- Simplified management

33. Classes of IPv4 Address

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation

34. Find the class of each address

- 00000010 10000010 01010100 01010101 : First bit is 0. Thus, class A
- 252.5.15.111 : class E
- 172.15.165.2 : class B

35. Differentiate IPV4 and IPV6

IPV4	IPV6
Header size : 20-60 bytes	Header size : 40bytes
Address length :32 bits	Address length :128 bits
Checksum field included	No Checksum field included
Version 4	Version 6
Address representation is in decimal format	Address representation is in hexadecimal format
Address consist of 4 fields separated by dot	Address consist of 8 fields separated by colon
Addresses divided into 5 classes, A B C D E	No classes
Eg : 128.90.12.1	2001:1A90:2123:1B23:2C13:1001:2123:3DEF

36. List applications of UDP and TCP

Ans: TCP : Text Communication, file transfer, email sending etc

UDP : Online gaming, Real-time multimedia streaming

37. List advantages and disadvantages of TCP

Ans:

Advantages of TCP

It is a reliable protocol.

It provides an error-checking mechanism as well as one for recovery.

It makes sure that the data reaches the proper destination in the exact order that it was sent.

Disadvantages of TCP

TCP is slower than UDP especially at the beginning of a file transfer. TCP's congestion control can also slow down data transfer to avoid n/w congestion

Additional expense to create and maintain connections since it is a connection oriented protocol.

TCP's use of ACKs and retransmissions can add latency which can affect real-time applications

Not suited for data transfers that require faster speed.

TCP uses more bandwidth than UDP, especially when establishing a connection,sending ACKs and retransmitting data

38. List advantages and disadvantages of UDP

Ans:

Advantages of UDP

Speed: UDP is faster than TCP because it does not have the overhead of establishing a connection and ensuring reliable data delivery.

Lower latency: Since there is no connection establishment, there is lower latency and faster response time.

Simplicity: UDP has a simpler protocol design than TCP, making it easier to implement and manage.

Broadcast support: UDP supports broadcasting to multiple recipients, making it useful for applications such as video streaming and online gaming.

UDP uses smaller packet sizes than TCP, which can reduce network congestion and improve overall network performance.

Disadvantages of UDP

No reliability: UDP does not guarantee delivery of packets or order of delivery, which can lead to missing or duplicate data.

No congestion control: UDP does not have congestion control, which means that it can send packets at a rate that can cause network congestion.

Limited use cases: UDP is not suitable for applications that require reliable data delivery, such as email or file transfers, and is better suited for applications that can tolerate some data loss, such as video streaming or online gaming.

No guaranteed ordering of packets while delivering.

Not suitable for critical data : Not ideal for transferring important data where accuracy and completeness matters, such as financial transactions

39. Explain the TCP connection establishment phase.

Ans:

Three-Way Handshaking

STEP 1

The client sends the first segment (SYN segment) in which only the SYN flag is set. This segment is for synchronization of sequence Numbers.

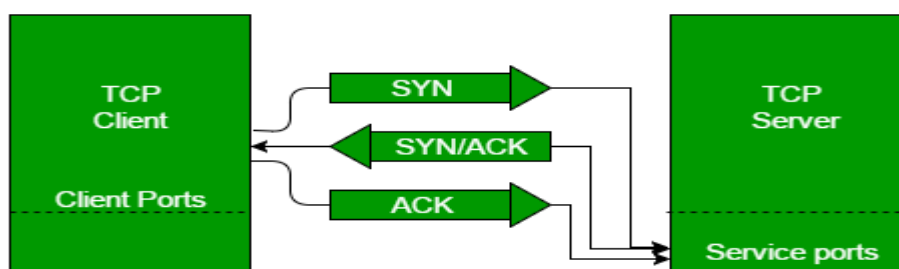
STEP 2

The server sends the second segment, a SYN + ACK segment with two flag bits set as: SYN and ACK.

STEP 3

The client sends the third segment. This is just an ACK segment.

It acknowledges the receipt of the second segment with the ACK flag and acknowledgment number field.



40. List any 6 differences between TCP and UDP

Ans:

TCP	UDP
Connection oriented	Connection less
Reliable as it guarentees the delivery of packet at receiver	Not reliable as it guarentees the delivery of packet at receiver
Slower	Faster
Retransmission of lost packets done	No retransmission of lost packets
TCP used by HTTP, FTP,SMTP etc	Used by DNS, DHCP, SNMP etc
Used where trustworthy communication is needed such as file transferring	Used where quick communication is necessary like live streaming

41. Explain the architecture of FTP

Ans: It is an application layer protocol. Used for transferring a file from one host to another

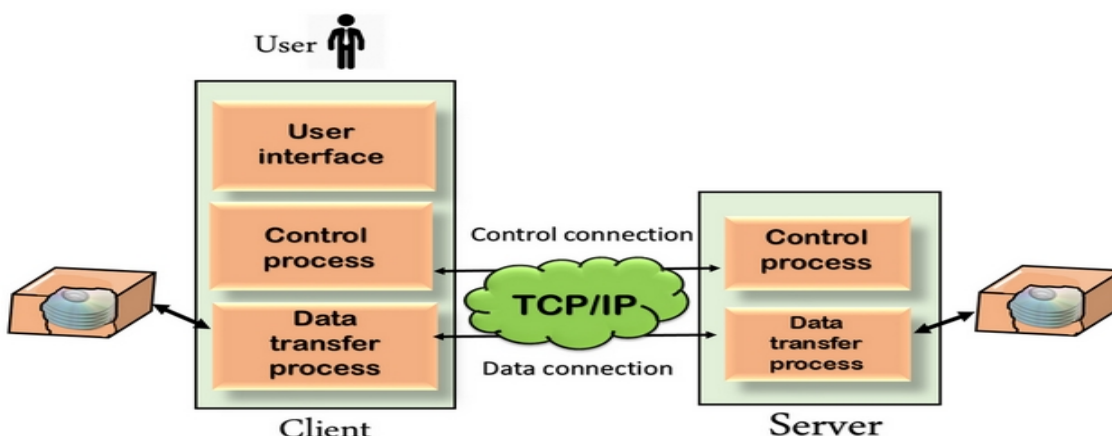
To transfer a file, FTP uses two TCP connections in parallel :

1. Control connection : for control information (commands and responses)
2. Data connection : Used for data transfer

Separation of commands and data transfer makes FTP more efficient.

FTP uses two well known TCP ports :

1. Port 21 is used for control connection.
2. Port 20 is used for data connection



42. Explain the purpose of DNS (Domain Name System).

Ans: It helps you find websites by translating easy-to-remember names (like www.example.com) into the numerical IP addresses (like 192.0.2.1) that computers use to locate each other on the internet.

43. Explain Resolution

Ans: Mapping a name to an address or address to a name is called resolution

1. recursive resolution

The client(resolver) can ask for a recursive answer from a name server.

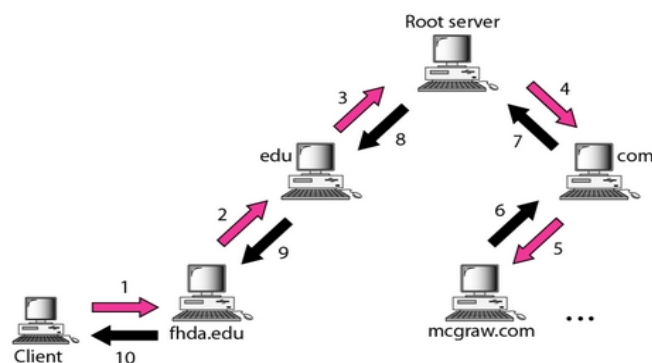
This means that the resolver expects the name server to supply the final answer.

If the server is the authority for the domain name, it checks its database and responds.

If the server is not the authority, it sends the request to another server (the parent usually) and waits for the response.

If the parent is the authority, it responds. otherwise it sends the query to yet another server.

When the query is finally resolved, the response travels back until it finally reaches the requesting client. This is called recursive resolution



2. Iterative resolution

If the client doesn't ask for a recursive answer, the mapping can be done iteratively

If the server is the authority for the name, it sends an answer.

If it is not, it returns to the client, the IP address of the server that it thinks can resolve the query.

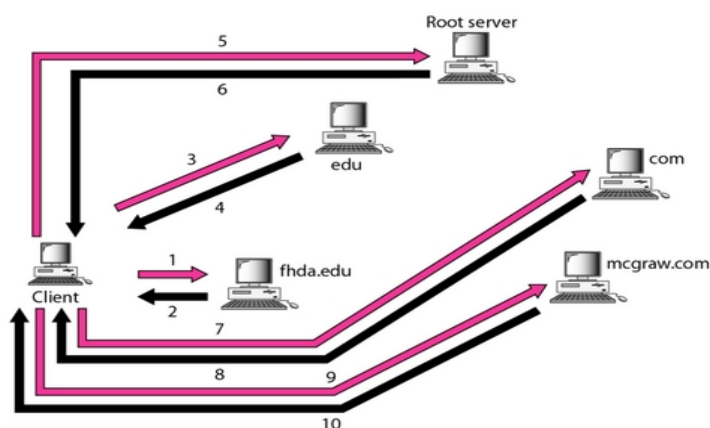
The client is responsible for repeating the query to this second server.

If the newly addressed server can resolve the problem, it answers the query with the IP address

Otherwise it returns the IP address of a new server to the client.

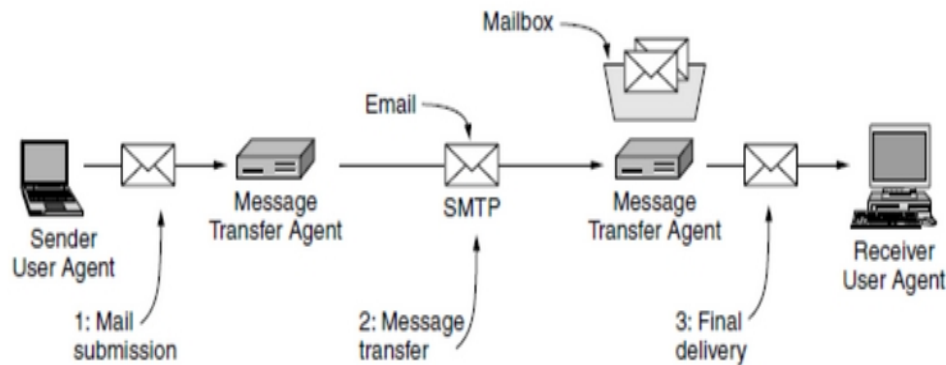
Now the client must repeat the query to the third server.

This process is called iterative resolution because the client repeats the same query to multiple servers.



44. Working of Email.

Electronic mail - Architecture



45. POP3 Vs IMAP

Ans:

Post Office Protocol (POP3)	Internet Message Access Protocol (IMAP)
<u>POP</u> is a simple protocol that only allows downloading messages from your Inbox to your local computer.	<u>IMAP (Internet Message Access Protocol)</u> is much more advanced and allows the user to see all the folders on the mail server.
The POP server listens on port 110, and the POP with SSL secure(POP3DS) server listens on port 995	The IMAP server listens on port 143, and the IMAP with SSL secure(IMAPDS) server listens on port 993.
In POP3 the mail can only be accessed from a single device at a time.	Messages can be accessed across multiple devices
To read the mail it has to be downloaded on the local system.	The mail content can be read partially before downloading.
The user can not organize mail in the mailbox of the mail server.	On the mail server, the user can directly arrange the email.
The user can not create, delete, or rename email on the mail server.	The user can create, delete, or rename an email on the mail server.
It is unidirectional i.e. all the changes made on a device do not affect the content present on the server.	It is Bi-directional i.e. all the changes made on the server or device are made on the other side too.

46. Explain the functions of each layer in OSI model (most important)

Ans:

1) Physical layer

- transmit the individual bits from one node to another node.
- Bit synchronisation
- **Line Configuration:** It defines the way how two or more devices can be connected physically.

- **Data transmission** : It defines the transmission mode whether it is simplex, half-duplex or full-duplex mode between the two devices on the network.
- **Topology** : It defines the way how network devices are arranged.
- **Signals**: It determines the type of the signal used for transmitting the information.

2) Data-Link Layer

- **Framing**: The data link layer translates the physical's raw bit stream into packets known as Frames.
- **Physical Addressing**
- **Flow Control**
- **Error Control** : error-free transfer of data frames.
- **Access Control**

3).Network Layer:

- Logical addressing
- Routing
- Packetising : A Network Layer receives the data from the upper layer and converts them into packets

4) Transport Layer

- **Service-point addressing**: Computers run several programs simultaneously. Due to this reason, the transmission of data from source to the destination not only from one computer to another computer but also from one process to another process. The transport layer adds the header that contains the address known as a service-point address or **port address**.
- **Segmentation and reassembly**: When the transport layer receives the message from the upper layer, it divides the message into multiple segments, and each segment is assigned with a sequence number that uniquely identifies each segment. When the message has arrived at the destination, then the transport layer reassembles the message based on their sequence numbers.
- Flow control
- Error control

5) Session Layer

- **Dialog control**: Session layer acts as a dialog controller that creates a dialog between two processes or we can say that it allows the communication between two processes which can be either half-duplex or full-duplex.
- **Synchronization**: Session layer adds some checkpoints when transmitting the data in a sequence. If some error occurs in the middle of the transmission of data, then the transmission will take place again from the checkpoint. This process is known as Synchronization and recovery.

6) Presentation Layer

- **Translation**: It converts the data from sender-dependent format into a common format and changes the common format into receiver-dependent format at the receiving end.

- **Encryption:** converting the sender-transmitted information into another form and sends the resulting message over the network.
- **Compression:** it reduces the number of bits to be transmitted. Data compression is very important in multimedia such as text, audio, video.

7) Application Layer

- An application layer serves as a window for users and application processes to access network service.
- This layer provides the network services to the end-users.
- An application layer allows a user to access the files in a remote computer, to retrieve the files from a computer and to manage the files in a remote computer.
- **Mail services:** An application layer provides the facility for email forwarding and storage.